

Assignment 1

Q1.

→ Conventional Energy

Non-Conventional Energy

1) Traditional Sources used for a long time

Newer & alternative sources developed recently.

2) Coal, Petroleum, Natural Gas, Hydropower

Solar, Wind, Biomass, Tidal Geothermal.

3) Limited & exhaustible

Abundant and naturally replenished

4) High Pollution

Low or almost no pollution

5) Lower initial cost but high fuel cost

Higher initial cost but low operating cost.

Q2.

→ Renewable Energy

Non-renewable Energy

1) Energy sources that can be replenished naturally

Energy sources that get depleted once used.

2) Solar, wind, Hydro, Biomass, Geothermal

Coal, Oil, Natural gas, Nuclear fuels.

3) Unlimited

Limited

4) Clean energy & less harmful

Cause pollution & greenhouse gases

5) High setup cost but free fuel

Lower setup cost but expensive fuel.

Q3.

→ Solar PV System	Solar Thermal System
1) Converts light into electricity	Converts sunlight into heat energy.
2) Electrical energy (DC)	Thermal energy
3) Solar PV cells	Solar collectors, heat exchange.
4) Lighting, solar pumps rooftop, power generation	Water heating, industrial heating.
5) Moderate efficiency 15% - 25%	Higher:- 30 - 70%
6) Low maintenance	Moderate Maintenance.
7) Higher cost per unit	Cost effective for heating applications.